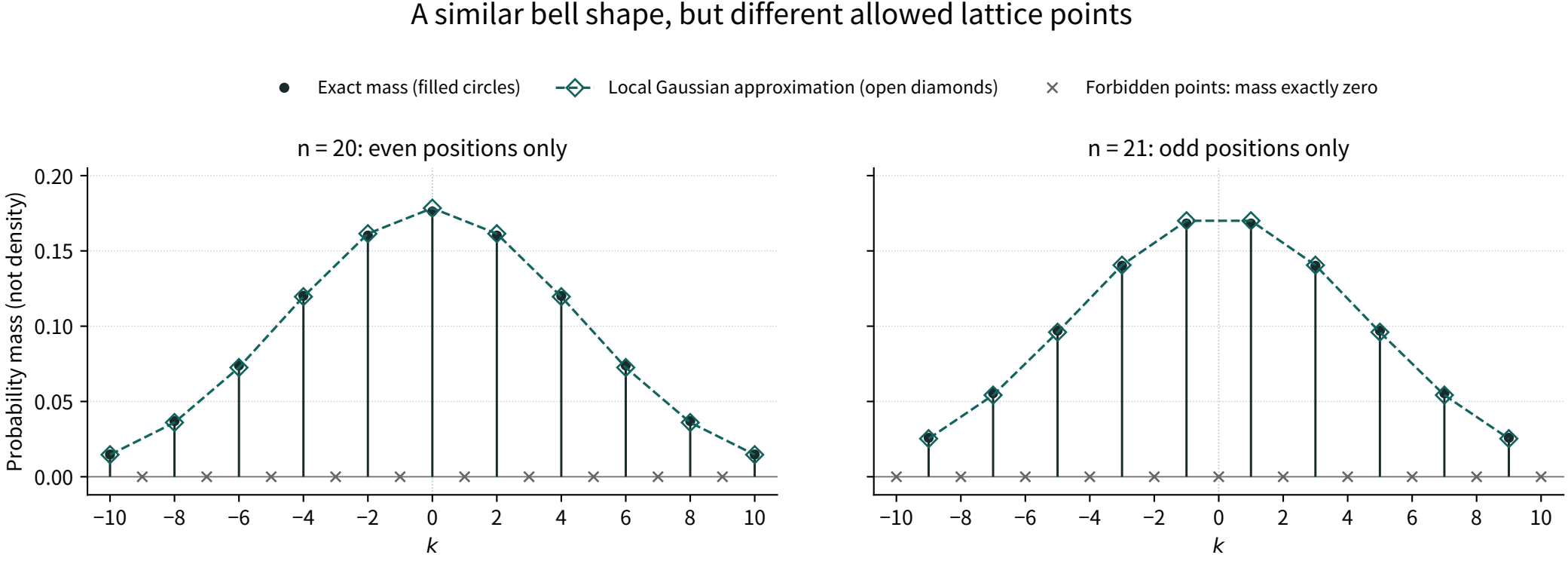
A similar bell shape, but different allowed lattice points
Exact mass (filled circles)
Local Gaussian approximation (open diamonds)
Forbidden points: mass exactly zero
n = 20: even positions only
n = 21: odd positions only
Probability mass (not density)
0.20
0.15
0.10
0.05
0.00
−10
−8
−6
−4
−2
0
2
4
6
8
10
k
$P(S_n = k) = 2^{-n}\binom{n}{(n+k)/2}$ (allowed points only)
$h = 2,\ \sigma = 1:\quad P(S_n = k) \approx \frac{2}{\sqrt{n}}\,\varphi(k/\sqrt{n})$